LAB EXPERIMENT 4

Experiment 4: Finite-State Machine for English Plural Nouns

Aim

To implement a simple finite-state machine that generates plural forms of English nouns using basic spelling rules.

Algorithm

1. Start the program.
2. Define a function to generate the plural form of a noun.
3. If a noun ends in s, sh, ch, x, or z, add 'es'.
4. If a noun ends in a consonant followed by y, replace y with 'ies'.
5. Otherwise, add 's' to the noun.
6. Apply the function to a list of nouns and display the singular and plural forms.
7. Stop the program.

Code

```
def plural(noun):
    if noun.endswith(("s", "sh", "ch", "x", "z")):
        return noun + "es"
    elif (noun.endswith("y") and len(noun) > 1
          and noun[-2].lower() not in "aeiou"):
        return noun[:-1] + "ies"
    else:
        return noun + "s"

words = ["cat", "bus", "box", "baby", "dish"]

print("Singular -> Plural")
for word in words:
    print(word, "->", plural(word))
```

Output / Screenshot

```
def plural(noun):
    if noun.endswith(("s", "sh", "ch", "x", "z")):
        return noun + "es"
    elif noun.endswith("y") and noun[-2] not in "aeiou":
        return noun[:-1] + "ies"
    else:
        return noun + "s"

words = ["cat", "bus", "box", "baby", "dish"]

for word in words:
    print(word, "->", plural(word))
```

```
cat -> cats
bus -> buses
box -> boxes
baby -> babies
dish -> dishes
```

Result

Thus, a simple rule-based finite-state machine was implemented to generate plural forms for the given English nouns.